Scale-Across Networking

Tad Hofmeister *Google Cloud*

Yawei Yin *Microsoft Azure*

Jeff Rahn *Meta*

Xiguang - Henry Wu *Broadcom*

Vijay Vusirikala *Arista Networks*

Panel Discussion



Tad Hofmeister

Google Cloud



Jeff Rahn

Meta



Yawei Yin

Microsoft



Xiguang (Henry) Wu

Broadcom

Moderator



Vijay Vusirikala

Arista Networks

Tad Hofmeister

Caveat

The material in this presentation includes design concepts that are being investigated but may not reflect current or future committed Google Cloud data center designs.

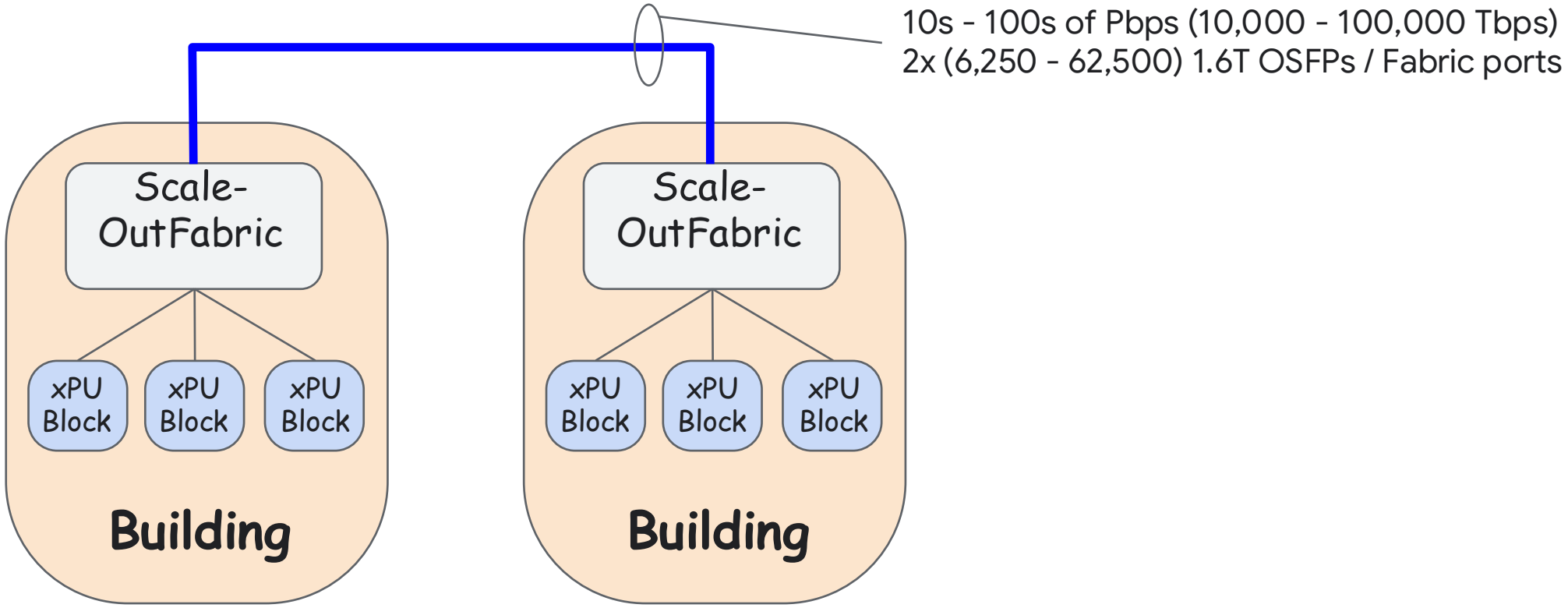
Example Data Center Campus

- New Albany data center campus in Central Ohio.
- Depending on building, power and/or space will limit the number of xPU nodes that can be deployed in a single building.
- Limit scale-out fabric (Virgo) to a single building.
- Inter-building scale out is required for training jobs that span multiple buildings.



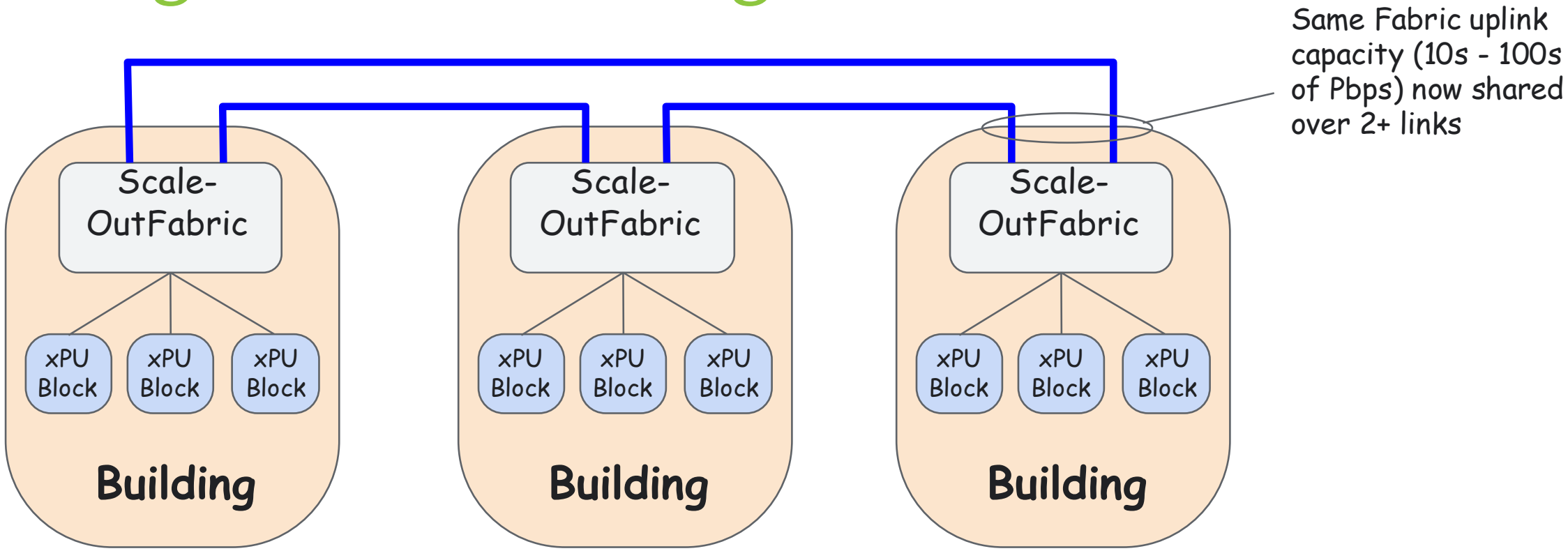
https://datacenters.google/discover-more/photo-gallery/

Point to Point Inter-Building Scale-Out



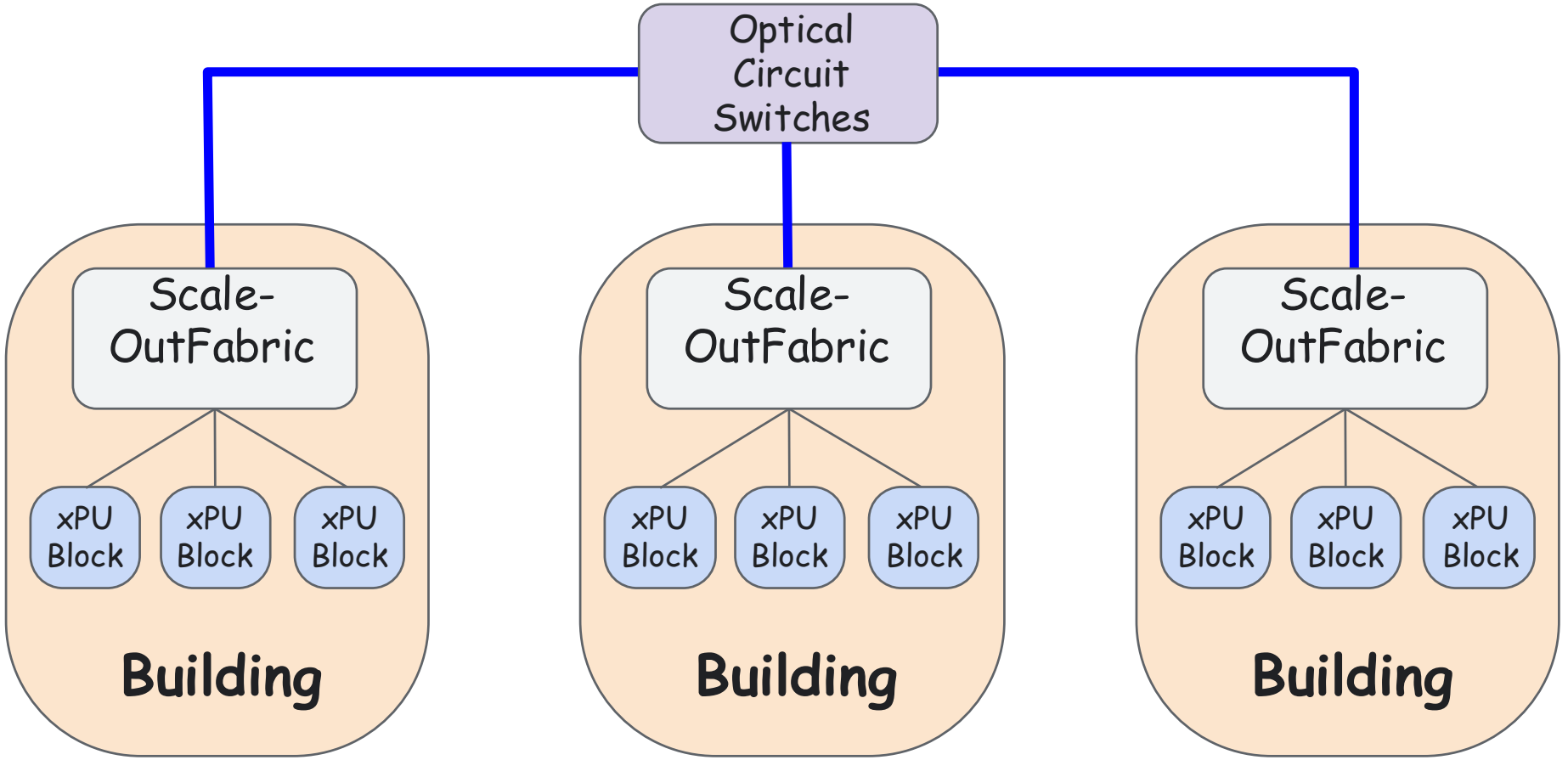
- Minimal over-subscription
  - 10s - 100s of Pbps BW between buildings
- Low Latency:
  - On average <2km fiber distance, but need to support outlier longer links
  - Direct connect fabric switch uplink ports

Scaling to More Buildings



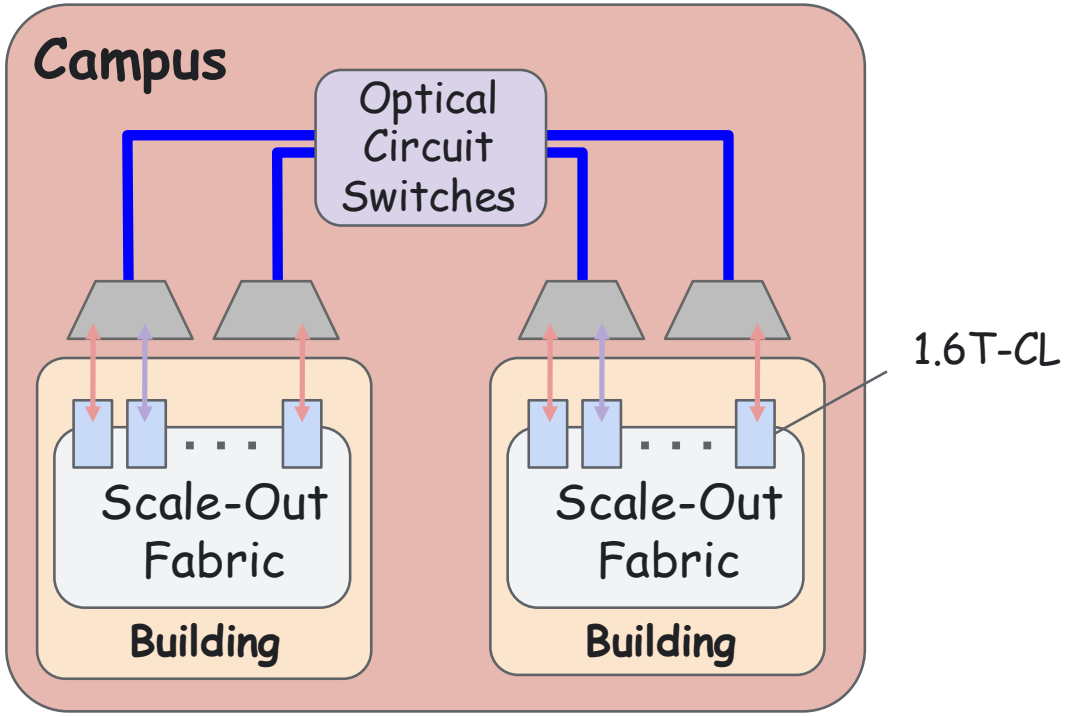
- With more buildings, fabric switch uplink BW must be shared for full mesh connection between fabrics.
  - ⇒ Oversubscription unless a campus spine fabric is added
- But an electronic switch campus spine adds: Latency, power, and cost.

Benefits of OCS



- OCS enables dynamic BW allocation between fabric adjacencies.
- Lower latency and lower cost than a spine layer electronic switch.

Coherent-Lite: A Better Option for Intra-Campus

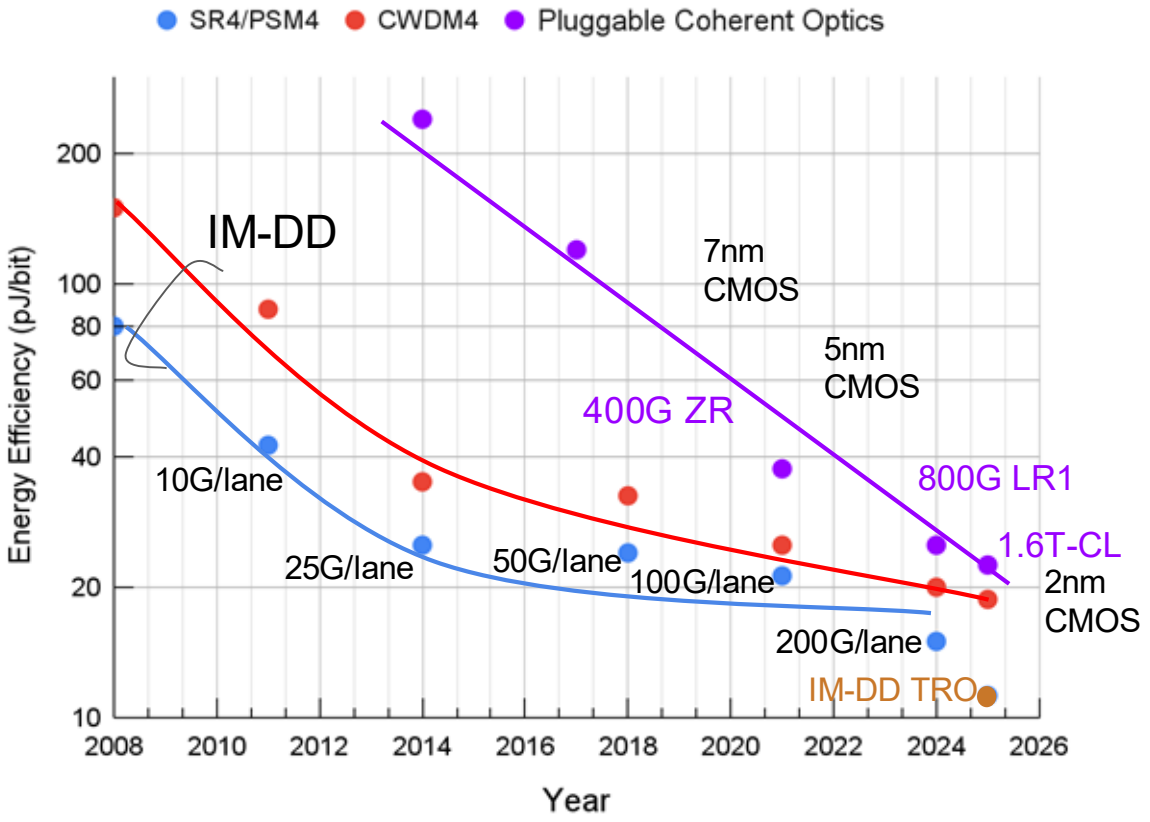


- Lower latency CL utilizes a lower latency FEC w.r.t. ZR.
- Lower power ⇒ full density on Scale-out fabric switches.
- O-band for lower dispersion.
- Why not IMDD?
  - 400G+ per channel IMDD has limited reach
  - Coherent-Lite (CL) is more spectrally efficient than IMDD

  ⇒ passive mux of 8, 16+ channels without managed line system chassis, fewer fibers and OCS ports are required

Coherent-Lite for Inter-Building Scale-Out

- 4x spectral efficiency compared to IM-DD systems
- Dispersion & nonlinearity tolerant
- Better link budget
  - Efficient optical modulators are required:
    - low loss, low Vpi (TFLN, Si-organic,…)
- Need to improve energy efficiency further
  - Low-power DSP (2nm CMOS) & low-latency FEC
  - O-band with few fixed wavelengths
- ⇒ OIF 1.6T-CL project
  - Low power, low latency interoperable specification



Ref: X. Zhou, OFC 2025, Tutorial, M4C.1

Conclusions

- Trends in number of xPU nodes and BW per node requires inter-building scale-out across the campus.
- BW and latency make direct fabric connectivity or via OCS attractive.
- 1.6T-CL (coherent lite) is desired:
  - Passive Optical Mux to combine O-band fixed wavelengths
    - Increase the BW per fiber pair
  - Low latency, low power for full density directly on fabric switches

Caveat: The material in this presentation includes design concepts that are being investigated but may not reflect current or future committed Google Cloud data center designs.

Call to Action

- Industry Ecosystem for Coherent-Lite
- Requirements and reference architecture for scale-across applications

Gigawatt scale clusters at Meta

Jeff Rahn

Meta Backbone & Edge Networks

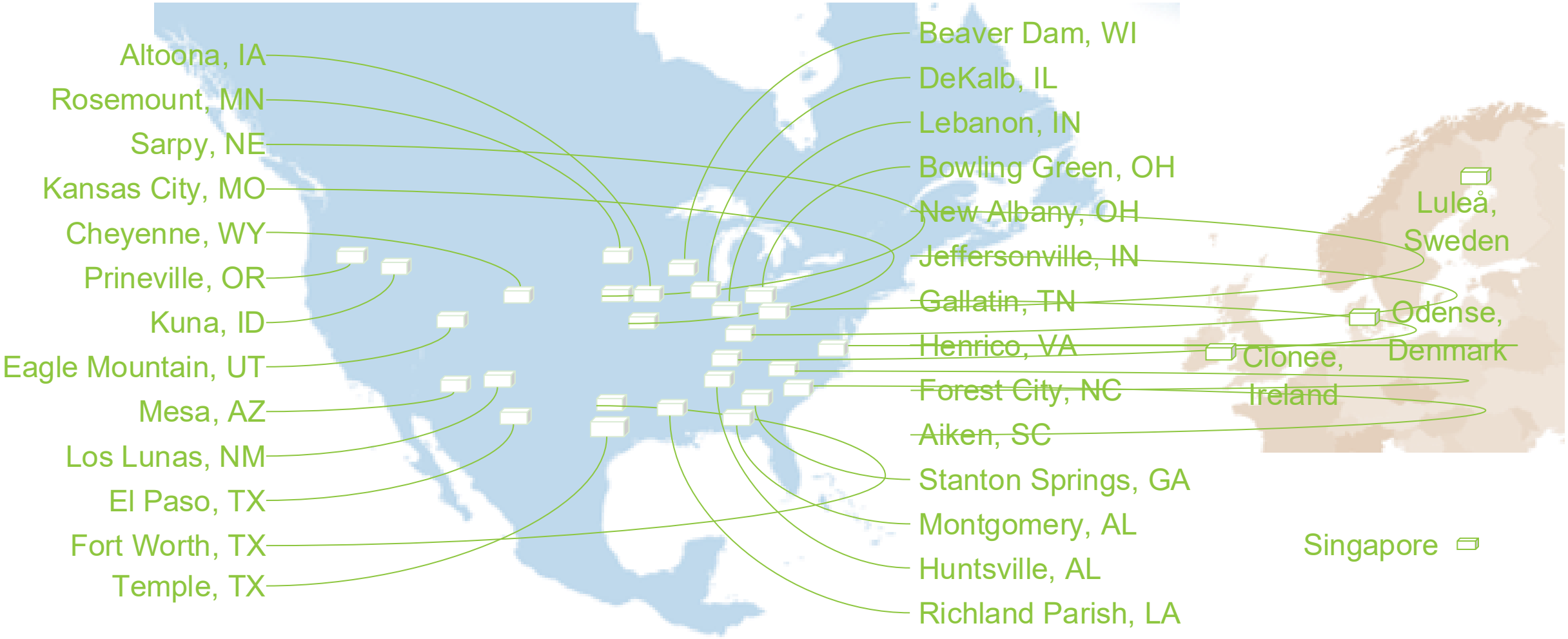


Outline

- Meta infrastructure and global backbone network
- How is our network evolving to enable gigawatt scale
- Optics enabling this evolution

Meta infrastructure and global backbone network

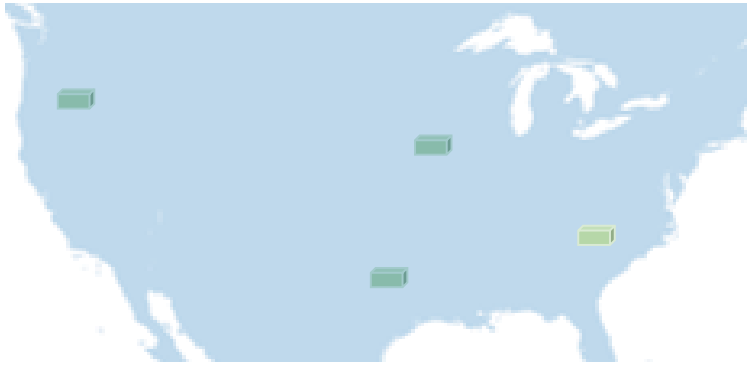
Meta Data Centers



Network Evolution

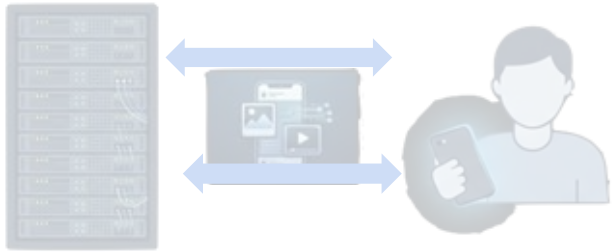
Data Center to User





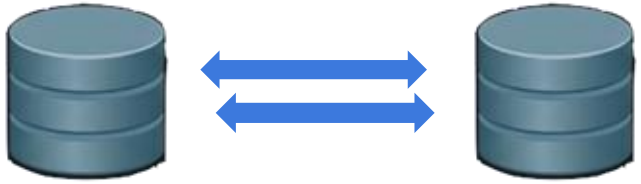
Network Evolution

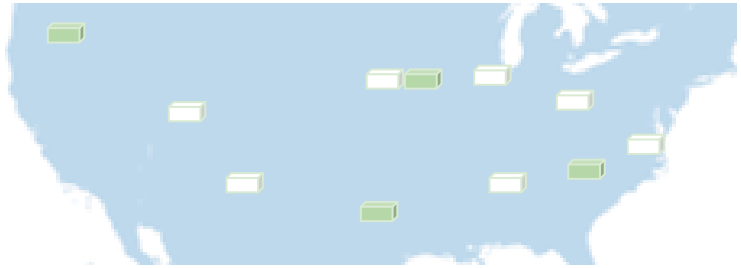
Data Center to User



Data Synchronization

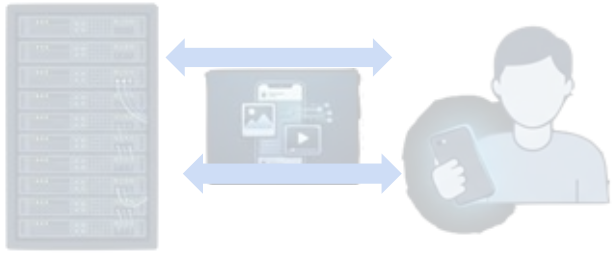
Critical data is on-site
DCs autonomous





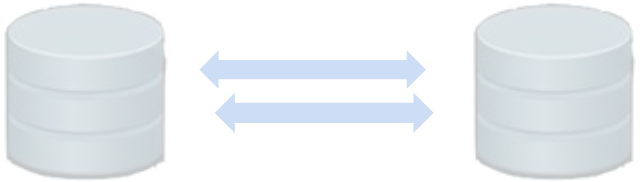
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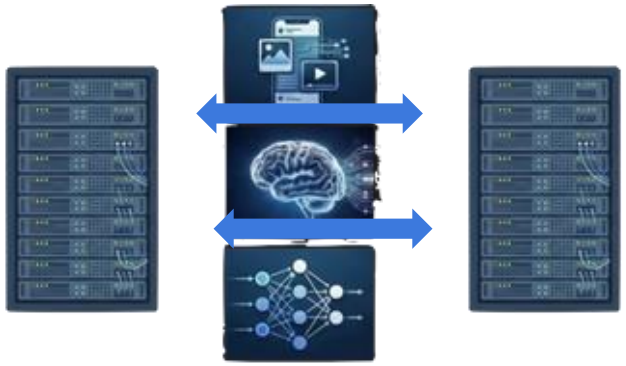
Data Synchronization

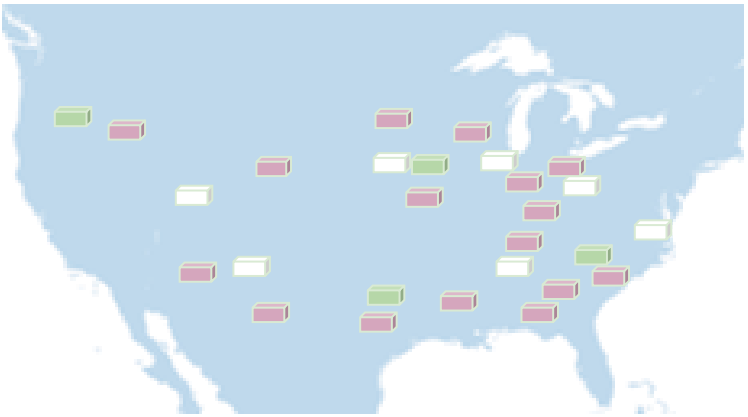
Critical data is on-site
DCs autonomous



Gigawatt scale clusters

Critical data available between sites
Geographically separate sites collaborate
Includes Scale-Across network

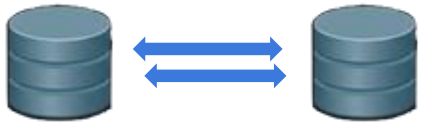




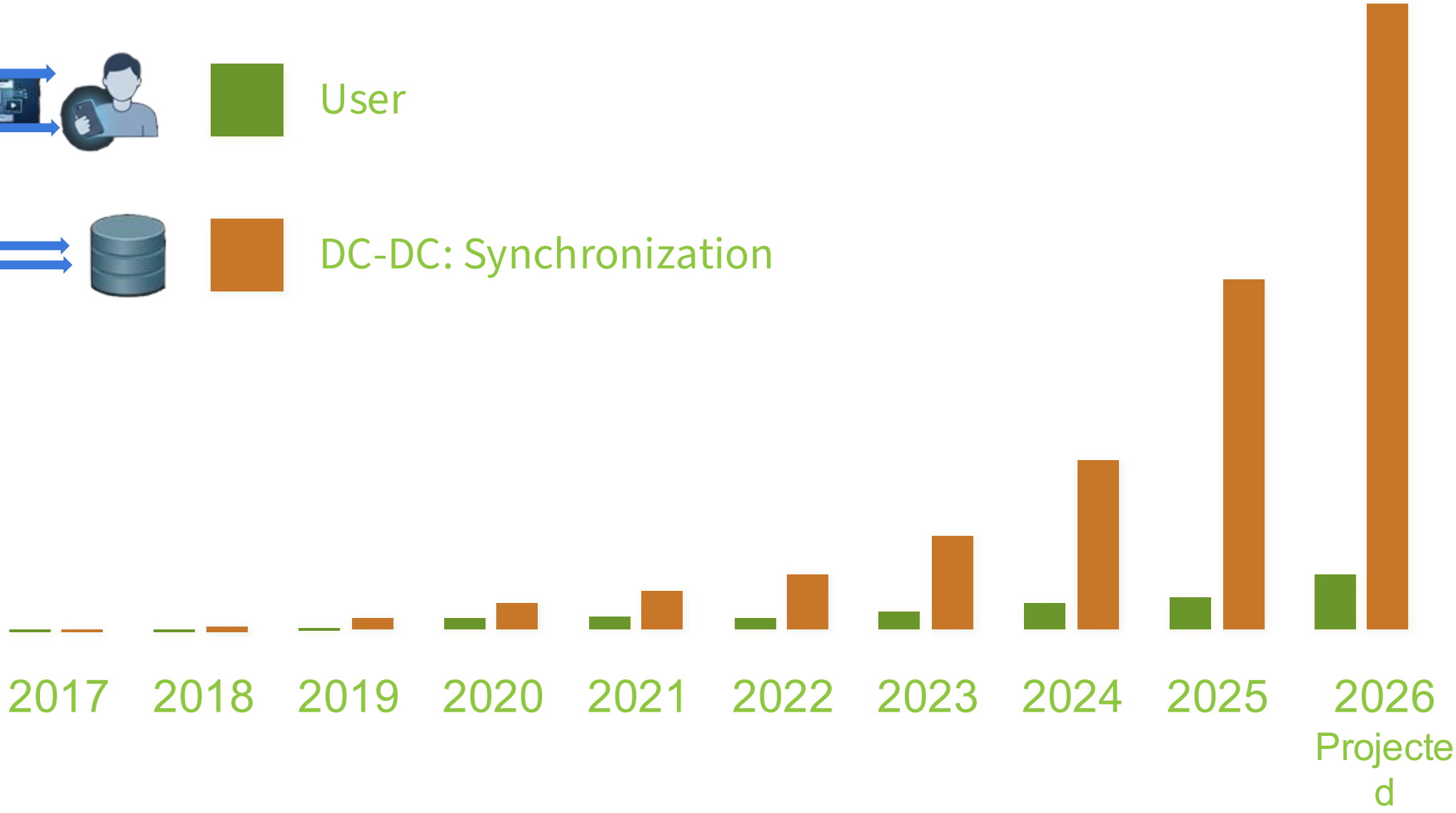
Network growth



User



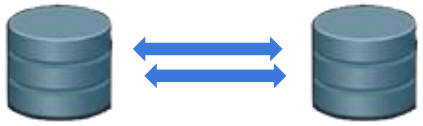
DC-DC: Synchronization



Network growth



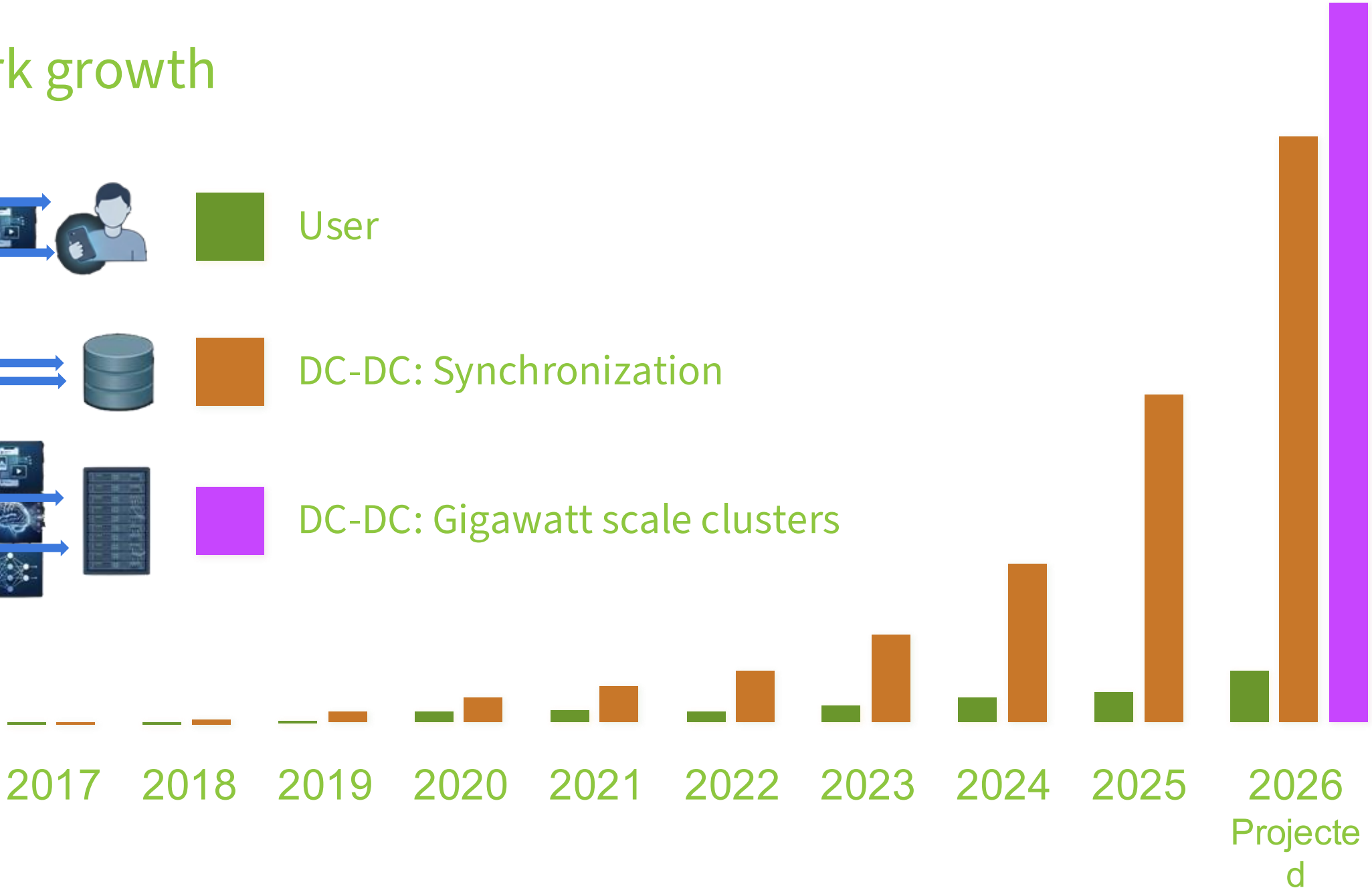
User



DC-DC: Synchronization

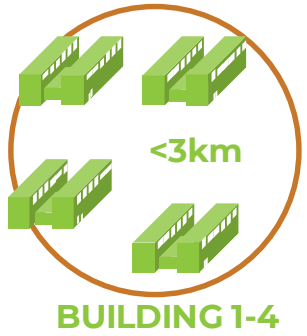


DC-DC: Gigawatt scale clusters



Evolution to gigawatt scale clusters

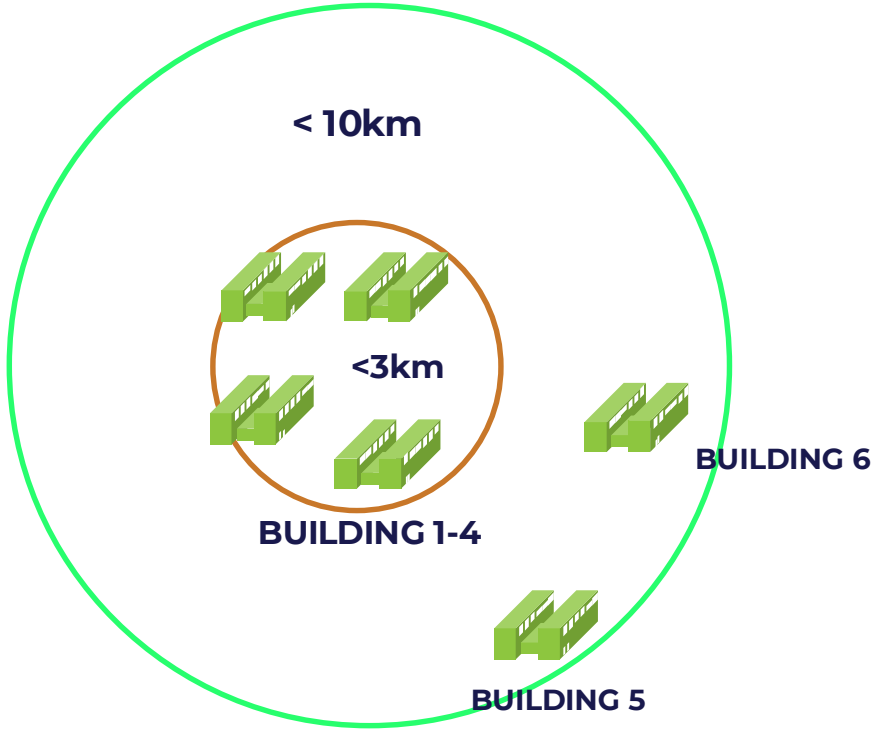
Meta has been deploying data center campuses within the capability of sort-reach optics



Evolution to gigawatt scale clusters

Meta has been deploying data center campuses within the capability of sort-reach optics

More recently, LR optics are used to further expand into larger clusters like Prometheus or across separate sites



Evolution to gigawatt scale clusters





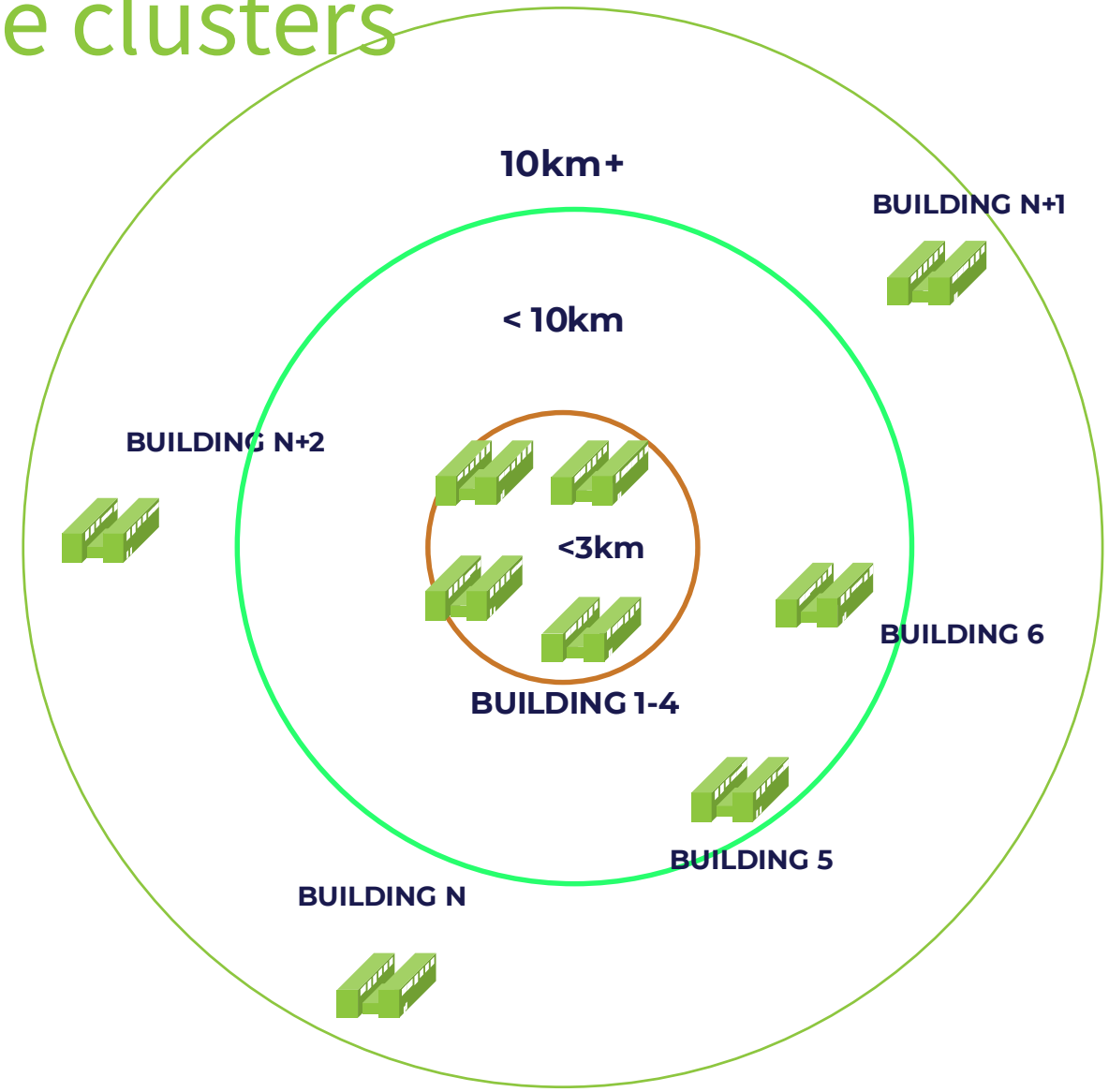


Evolution to gigawatt scale clusters

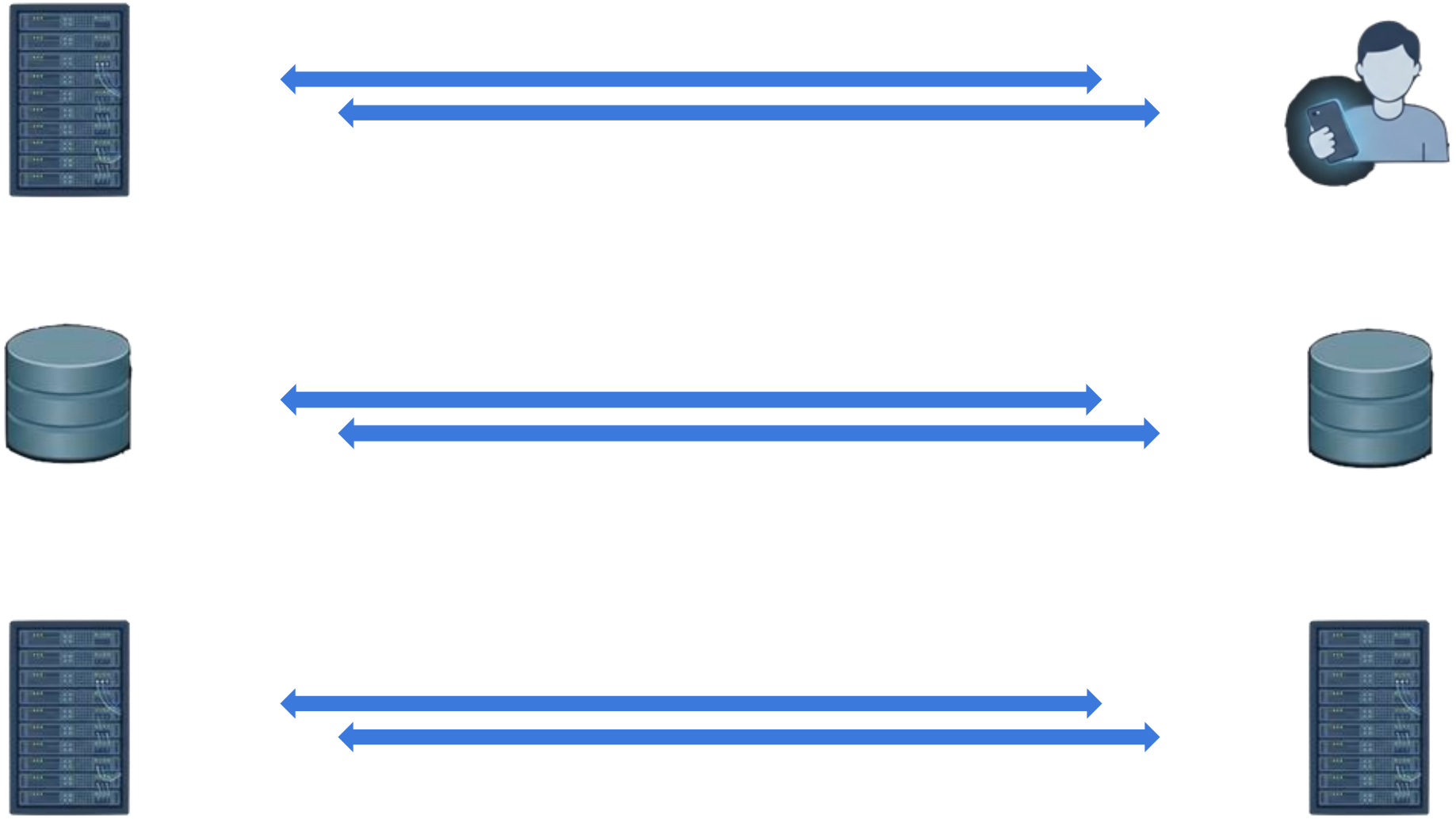
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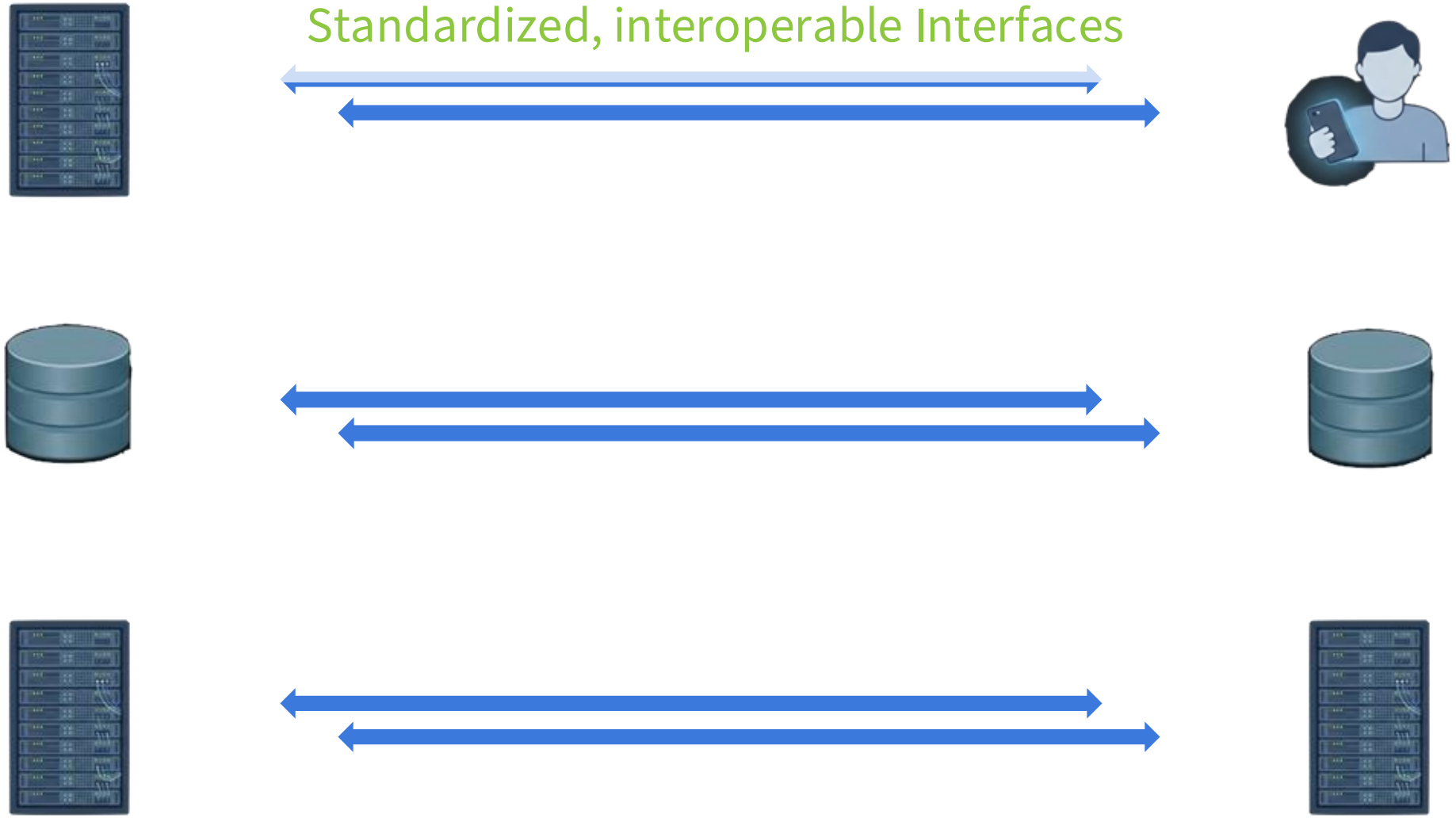
Further expansion to reach gigawatt scale can be achieved by leveraging 800ZR+ and DWDM



Optics ecosystem enabling Gigawatt scale



Optics ecosystem enabling Gigawatt scale



Optics ecosystem enabling Gigawatt scale



Standardized, interoperable Interfaces

Open line systems















Optics ecosystem enabling Gigawatt scale



Standardized, interoperable Interfaces

Open line systems



Generational Interoperability to support upgrades

400ZR ⇄ 800ZR ⇄ 1600ZR









Optics ecosystem enabling Gigawatt scale



Standardized, interoperable Interfaces

Open line systems



Generational Interoperability to support upgrades

400ZR ⇄ 800ZR ⇄ 1600ZR



Full fill of C and L bands for fiber efficiency







Optics ecosystem enabling Gigawatt scale



Standardized, interoperable Interfaces

Open line systems



Generational Interoperability to support upgrades

400ZR ⇄ 800ZR ⇄ 1600ZR



Full fill of C and L bands for fiber efficiency

Shared optical solutions between DC and backbone









Optics ecosystem enabling Gigawatt scale



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Optics fungible across applications







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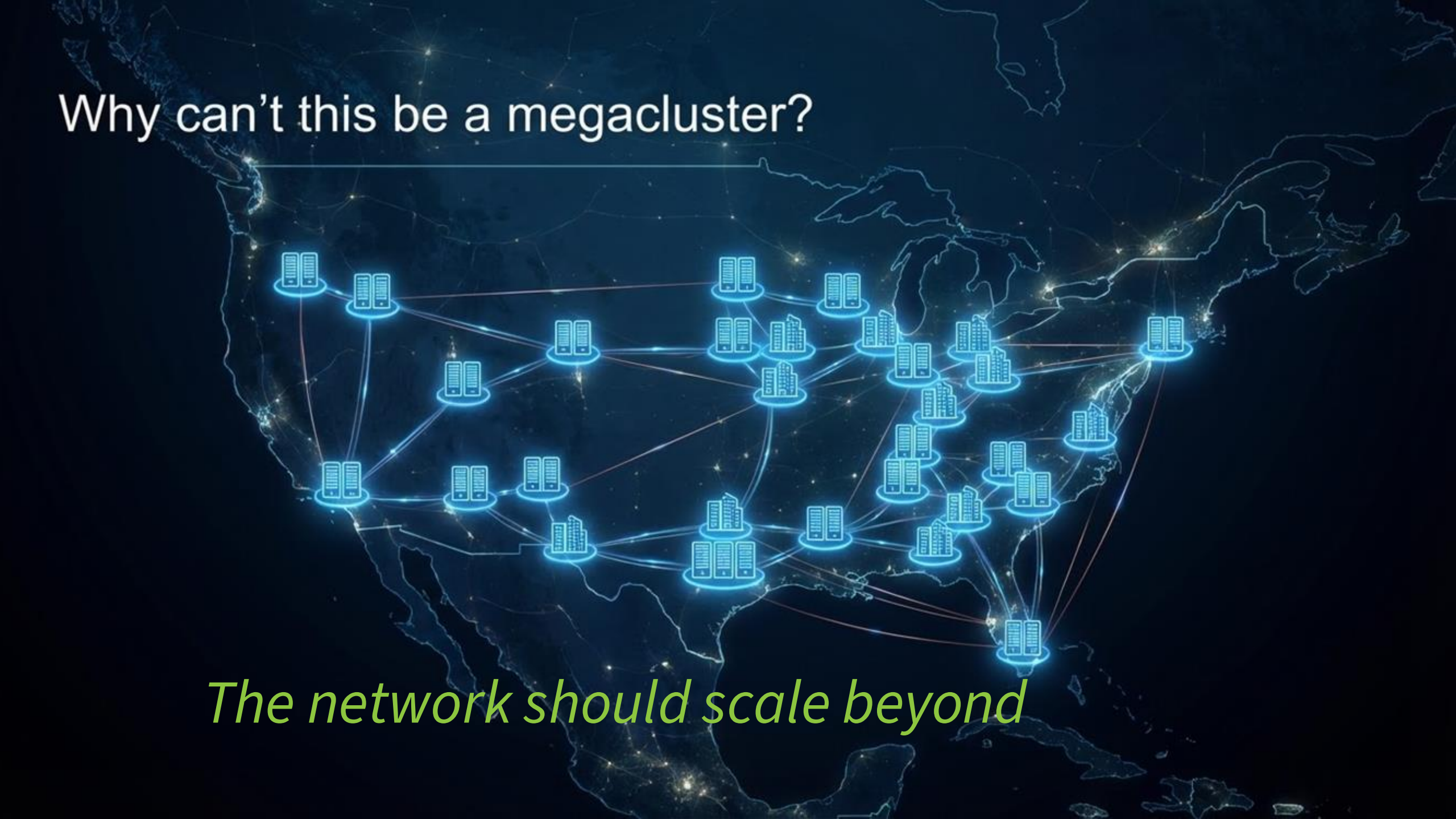
Optical link protection







Why can’t this be a megacluster?



The network should scale beyond

OCP SPECIAL FOCUS: PHOTONICS

Scaling Across With Optics: Building AI Ready Fabrics from IMDD Pluggable to Coherent DCI

Yawei Yin

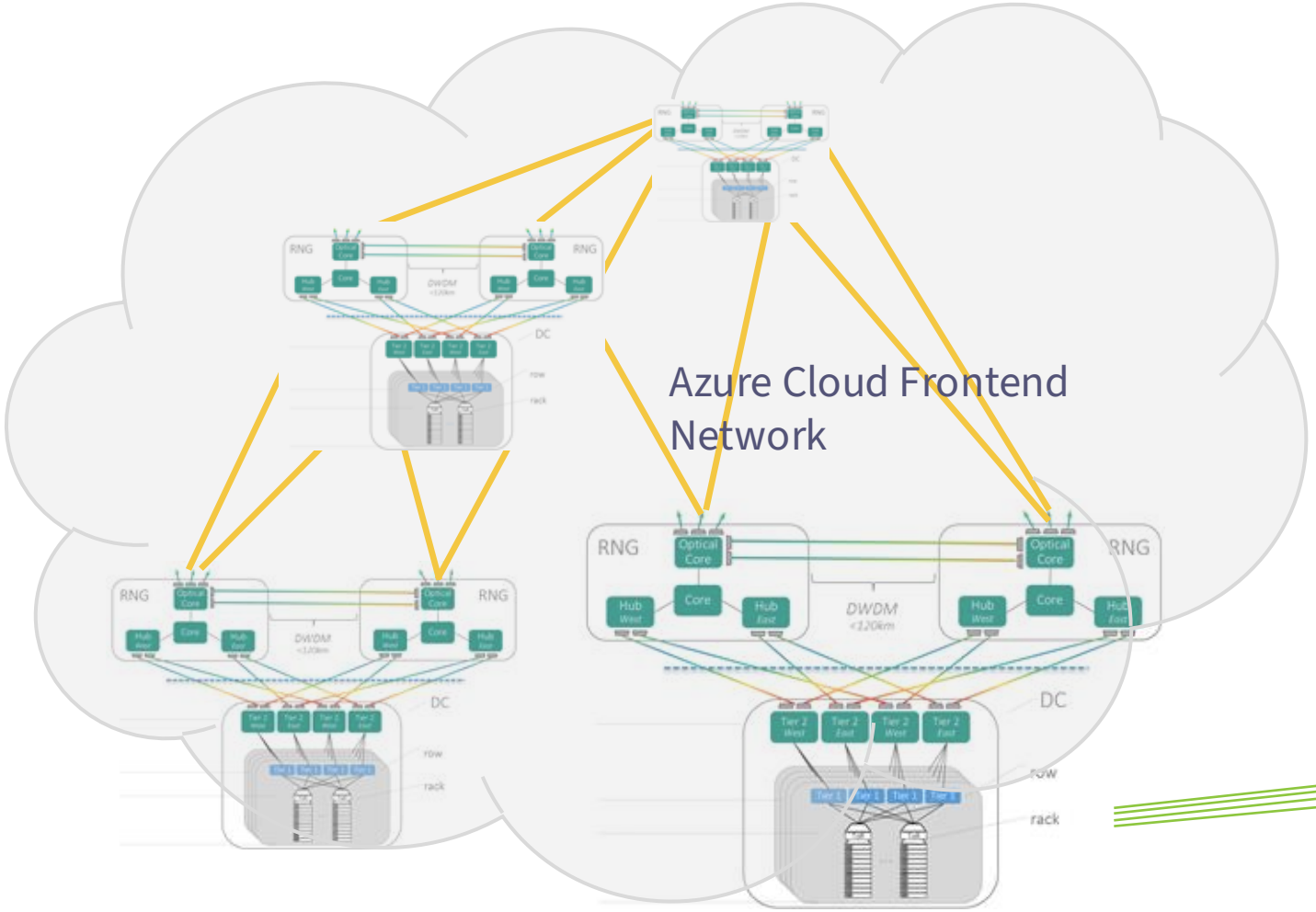
Principal Network Developer, Microsoft

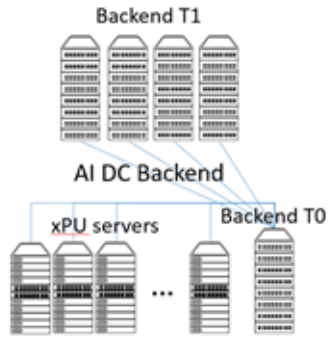
OCP PLATINUM Member

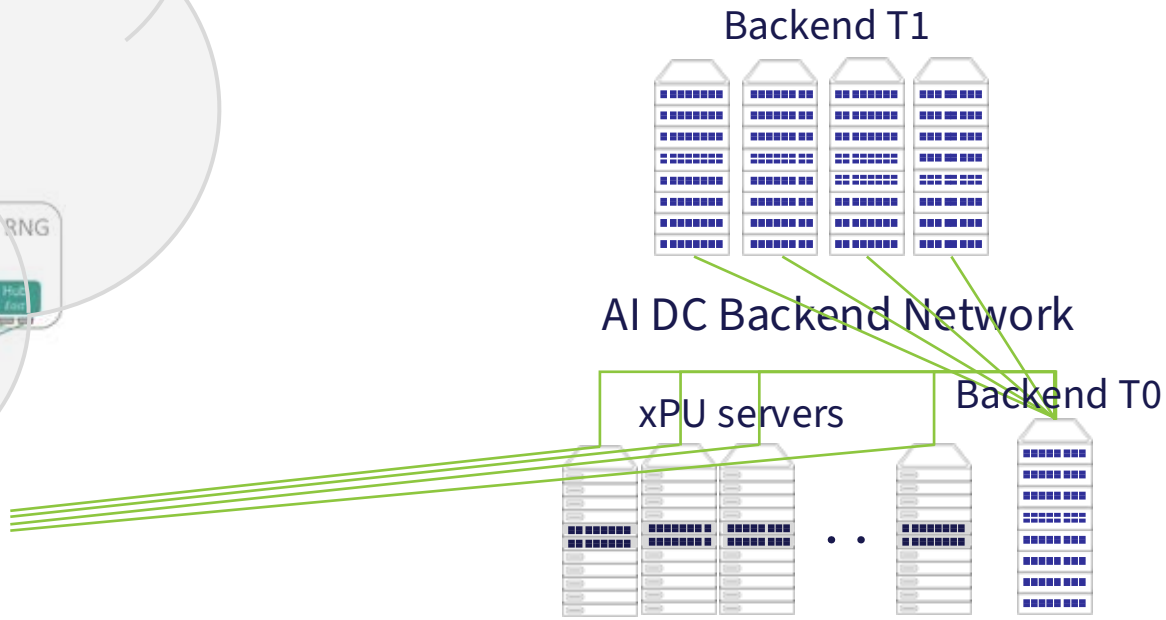
Agenda

- Is Scaling Across Needed by Training and Inference
  - Cloud and AI DCs: A Networking View
  - Scale up, scale out, and scale across
  - The universal need: Energy
- What are the limiting factors
  - Reliability
  - Energy Efficiency
  - Bandwidth, Latency and Jitter
- Enabling Technologies and Enabling Industry Alignments

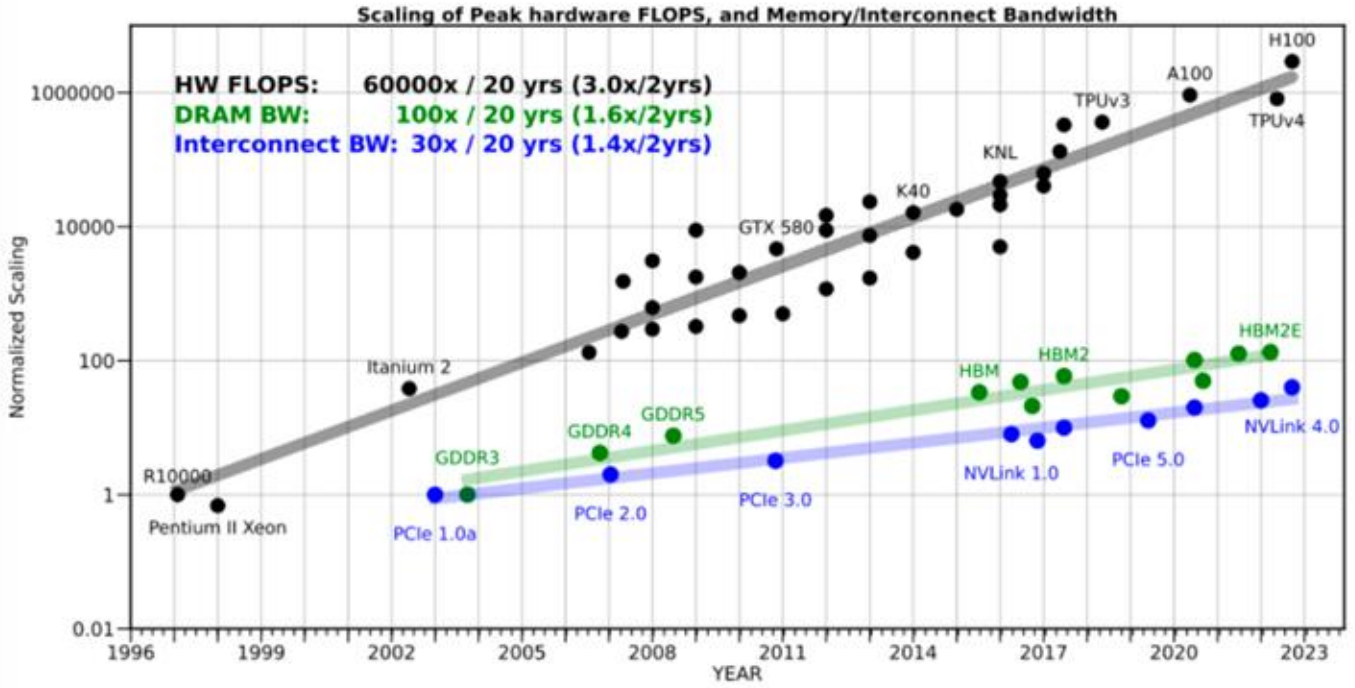
Cloud DC and AI DC @ Microsoft



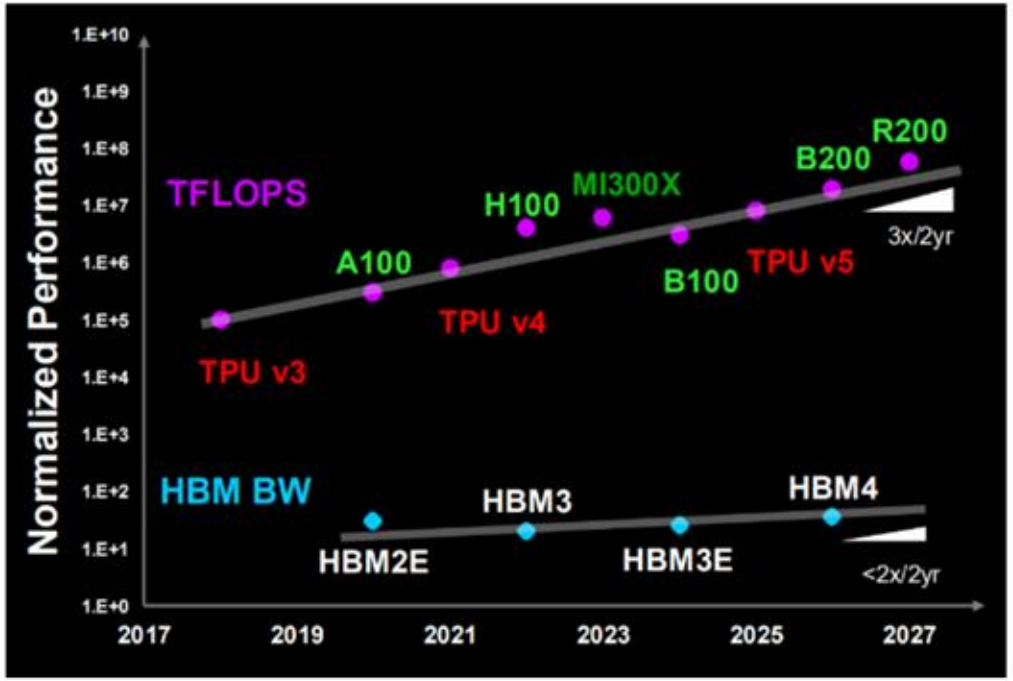




Scaling at different pace

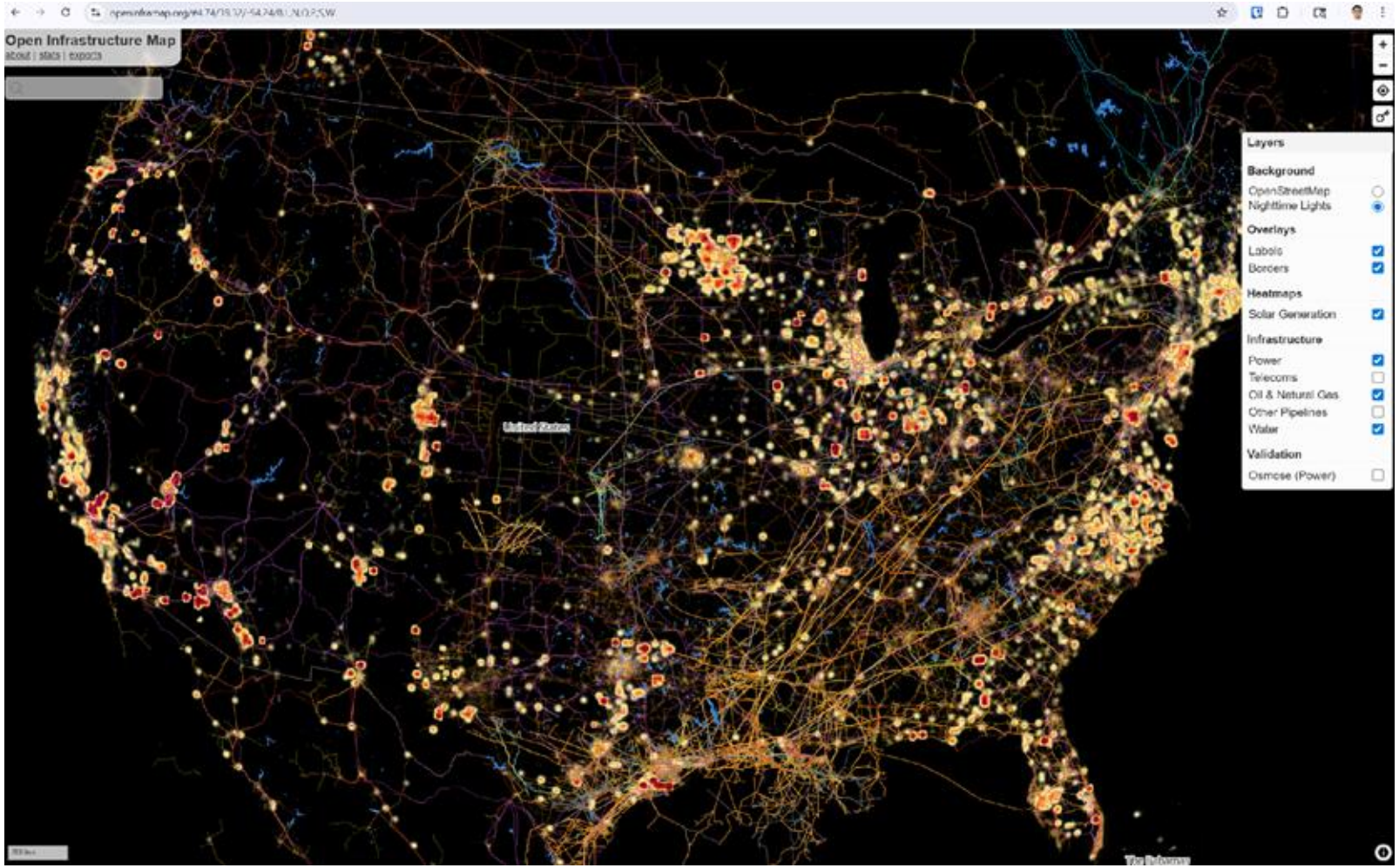


Source: "AI and Memory Wall", Amir Gholami, et., al.
https://arxiv.org/pdf/2403.14123



Source: R. Sreeramaneni, "Evolving Memory Architectures for AI," Micron, Hot Chips 2026, Aug. 2026

Scaling w/ Power Limitation



Reference: “Open Infrastructure Map”, Russ Garrett, from London, UK
https://openinframap.org/#4.74/39.32/-94.24/B,L,N,O,P,S,W

- Traditional single data center power capacity: 50 - 300 MW typical
- Next-gen training clusters demand 1 to 5+ GW
- Power availability is geographically distributed — no single site can deliver enough
- Renewable energy sources are location-dependent (solar, wind, hydro)
- Regulatory and grid interconnection timelines: 3–7 years for new sites

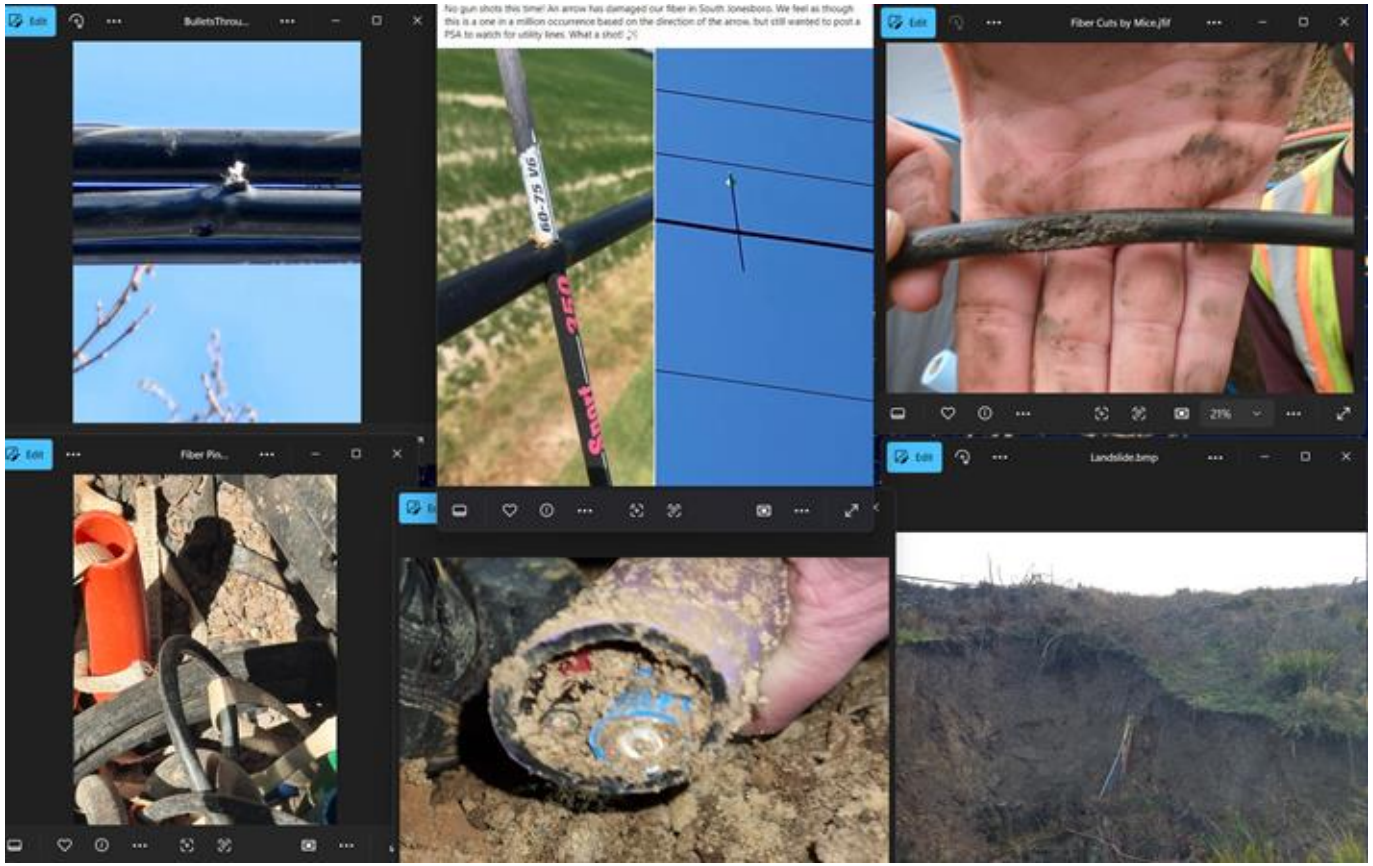
⚡ **Conclusion:**
Distributed multi-site training & inference is not optional — it is an architectural necessity for frontier AI models

Scaling w/ Power Efficiency

Technology	Energy Efficiency	Reach	Use Case	Status
Traditional Transponder (coherent pizza-box DWDM)	> 1000 (pj/bit)	> 1,000km	Long-haul DWDM, inter-region	Mature/Legacy
DCI Metro IP over DWDM (IMDD PAM4, Coherent 400ZR)	45-55 (pj/bit)	10 – 120km	Metro DCI DWDM, intra-region	Deployed/Mains tream
FRO (Fully-Retimed Optics, dual-DSP, 800G-1.6T)	18-25 (pj/bit)	500m-10km	Data center fabric	Deployed
Half-Retimed Optics (a.k.a. LRO / RTLR: retimed Tx, linear Rx)	10-18 (pj/bit)	500m-2km	Intra-DC, AI cluster fabric	Emerging
LPO - Linear Pluggable Optics ("unretimed" in module, no DSP)	7-10 (pj/bit)	500m+	Intra-DC, AI cluster fabric	Deploying
NPO - Near Packaged Optics (optical engine adjacent to ASIC)	5-8 (pj/bit)	500m+	Intra-DC, Scale-up interconnects	Emerging
CPO – Co-Packaged Optics (optical engine on ASIC substrate)	1-5 (pj/bit)	10-30m+	Scale-up interconnects, XPU/ASIC chip shoreline	Emerging

Other challenges with scale

- Reliability
  - OSP (Outside Plant) fibers' market availability is around 97.7%
  - Hardware FIT
  - Link Flaps (Link margin & burst errors)
- Traditional cloud computing are more tolerant to link failures (depending on the traffic types)
- AI clusters are tightly coupled super computers
  - Synchronous training jobs are not tolerant to failures
  - A single failure pauses the progress

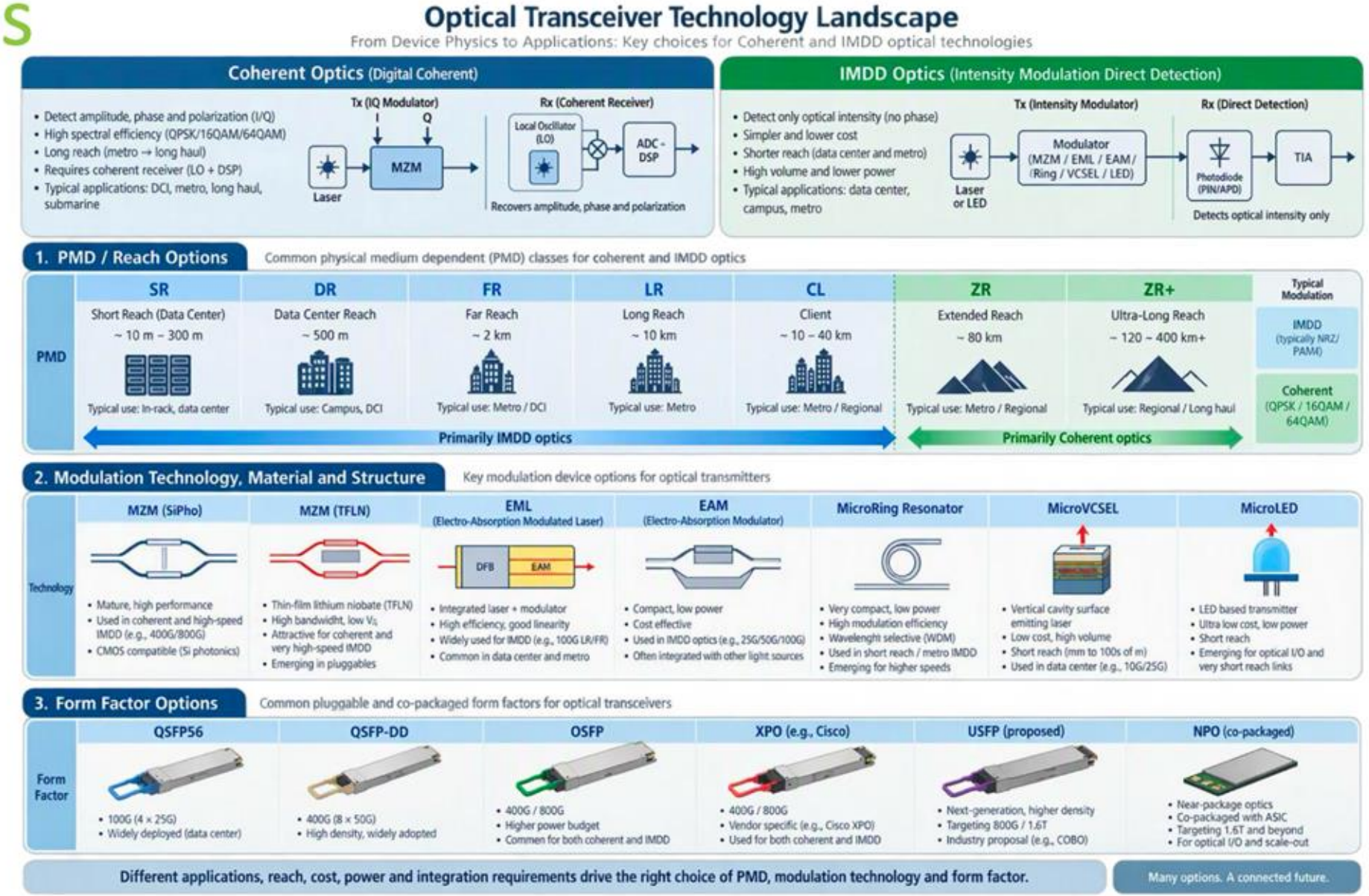


IMDD & Coherent

- IMDD use cases, and the projection into CPO
  - Public Cloud & AI Frontend (<2km) – IMDD transceivers
  - AI Backend Scale Out (<1km): Fast & Narrow, pluggable or NPO?
  - AI Backend Scale Up (<50m): Fast & Narrow v.s. Slow & Wide?
  - AI Backend Scale In (<1m): Chip to Chip & Chip to Memory
- Coherent use cases
  - Scale across distances (buildings, campuses, cities)
  - Scale across loss (OCS, patch panels, shuffle boxes, etc.)
- Tradeoff of SKU management and cost/power efficiency
  - Coherent: CL, ZR, ZR+, Transponders -> CL, ZR+ ?
  - IMDD: LR, FR, DR, SR, -> FR, DR?

Call to actions

- Less is more
  - Diverging requirements are creating R&D and manufacturing resource misspend
  - Encourage Innovation,
  - limit asks to mass production variants



[TRACK NAME]

The Switch For Scale-across

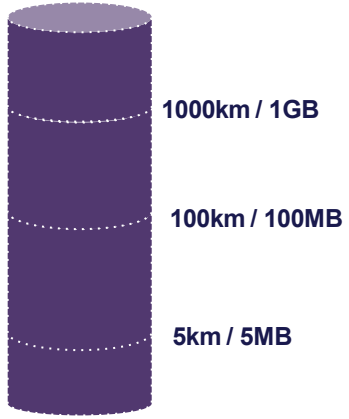
Henry (Xiguang) Wu

[Product Management, Core Switch Group, Broadcom]

Scale-Across Challenges

Bandwidth Delay Product Explosion

800GE BDP In-flight traffic

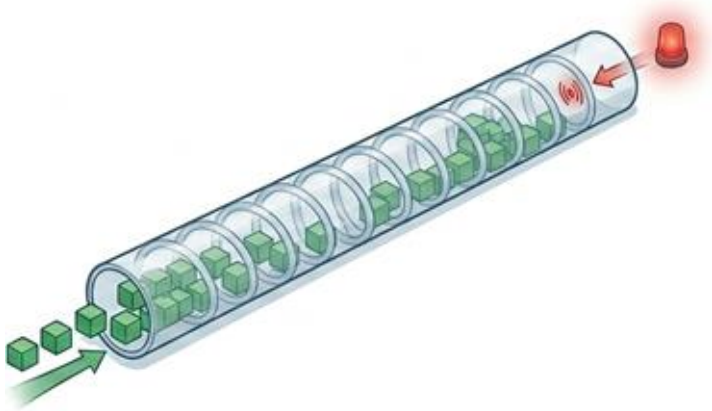


Scale-across AI fabric require: Large in-flight windows, congestion stabilization

BDP = Bandwidth × RTT

As RTT scales to milliseconds, BDP explodes

Long RTT Slows Feedback Loops

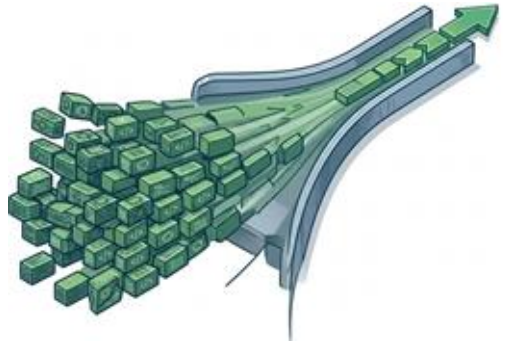


Congestion signaling requires time to travel

Packets continue to arrive

Congestion escalates rapidly

Utilization Efficiency Matters



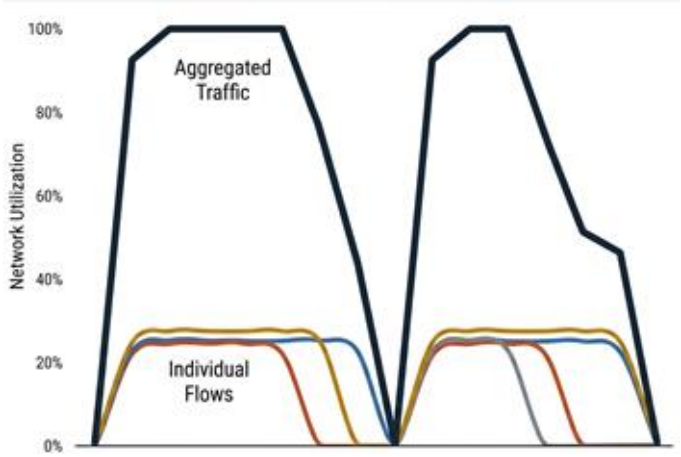
High-cost links demand peak performance.

Operators target:

- High sustained utilization
- Minimized idle cycles
- Maximum throughput efficiency

Scale-Across Challenges

Synchronized and Bursty Traffic



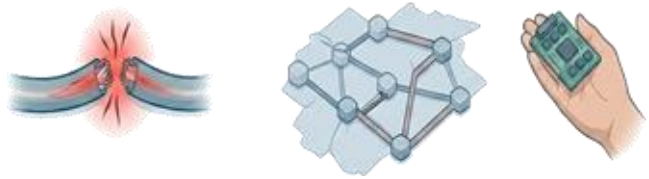
Many endpoints inject simultaneously
Fan-in/fan-out traffic spikes
Burst synchronization across clusters
Oversubscription at DCI boundaries

Massive Scale & Resource Demands



Large ECMP fan-out/in scales
Multi-million LPM/FIB lookup scale
Massive tunnel scale and formats

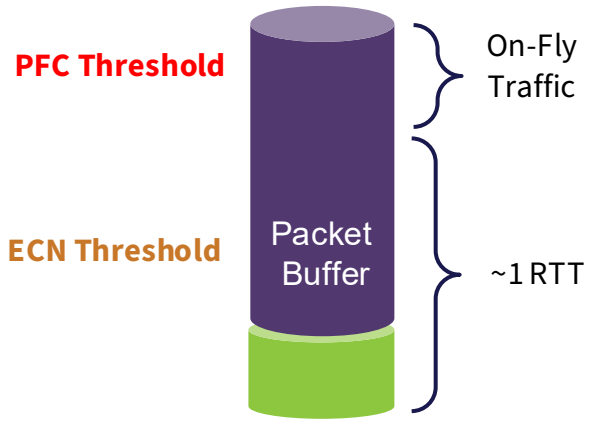
Failure and Reroute Resilience



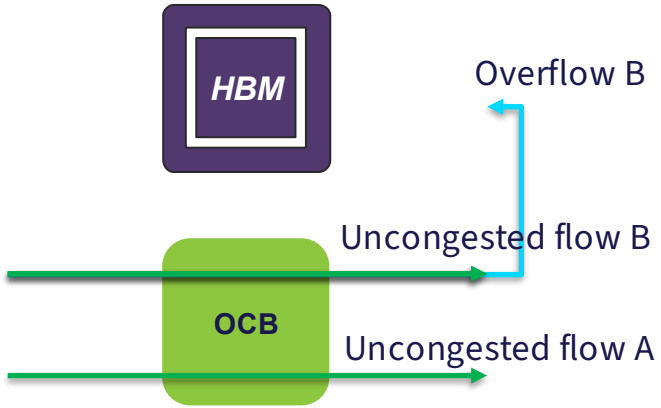
Infrastructure must maintain uptime despite:

- Optical failures and link degradation
- Route changes and circuit reconfiguration
- Scheduled maintenance windows

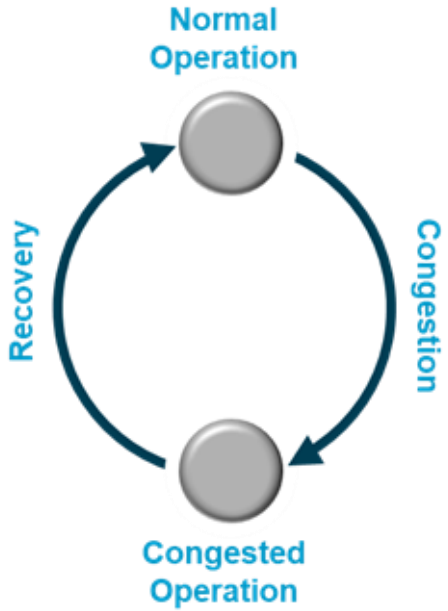
Key Enablers



Buffer Capacity



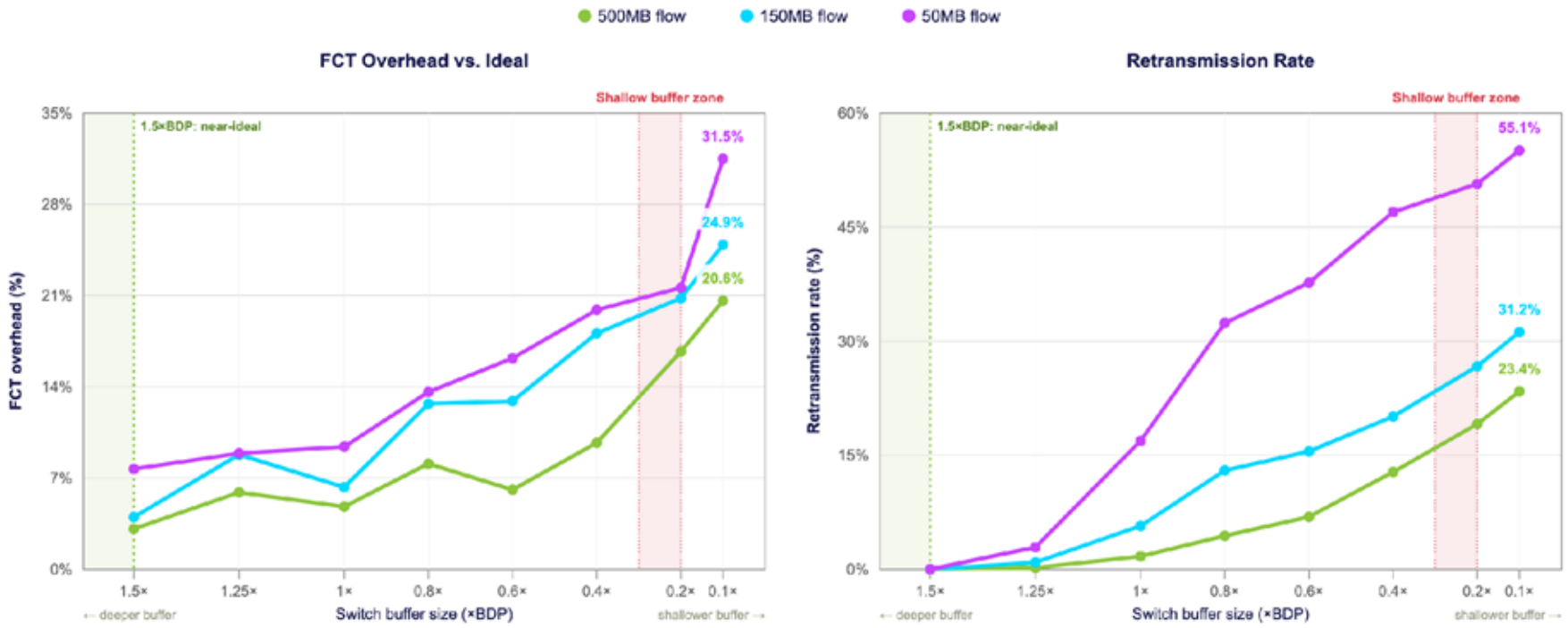
Traffic Management



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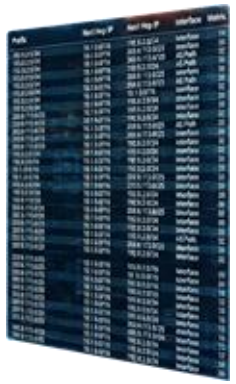
Buffer Depth Determines Completion Time

Across 50–500MB flows, ≥1.5×BDP keeps FCT near-ideal and retransmission at zero — larger messages soften, but don't remove, the need for adequate buffer



Message-size sweep run at 10km under NSCC. The ≥1.5×BDP threshold itself is confirmed at 10km, 30km, and 100km, under both NSCC and RCCC.

Key Enablers



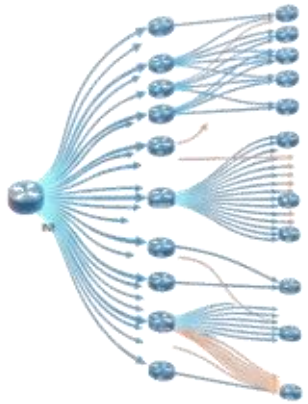
Modular Database

Millions of routes
IPv4 / IPv6
Flexible
Large-scale Tunneling



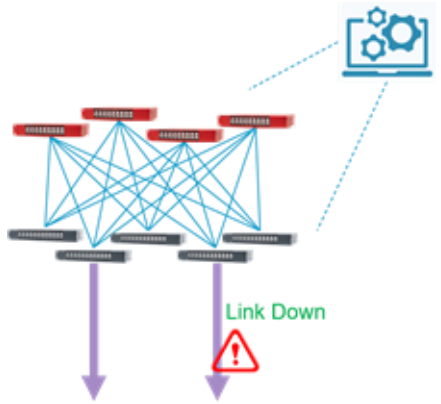
Streaming Telemetry

All-clusters-view
Head and tail editing
Congestion monitoring



Mega-Scale ECMP

Tens of thousands of ECMP groups
Thousands of group members



Protection Switching

React in uSec
Lossless recovery
Robust ECMP re-mapping



Line-rate Security

MACsec
IPSec
TunnelSec

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Key Takeaway: Headroom Beats Retransmission

The simulation isolates exactly how much buffer is enough — and shows the answer doesn't depend on distance or CC protocol

① Near-Ideal FCT

Buffer ≥1.5×BDP lands within a few percent of ideal — at every distance tested.

② Retransmission First

Below threshold, retransmission climbs before completion time does — then both grow sharply.

③ CC-Agnostic

The threshold holds under NSCC and RCCC alike — a buffer-sizing property, not a tuning trick.

At Switch Scale, Shallow Buffers Can't Get There

1.5×BDP at 100T switch throughput. Real buffers need even more — usable capacity always trails nominal, so shallow, on-chip-only buffers can't get there.

10km	30km	100km
~1.9GB	~5.6GB	~18.8GB


Thank
You!
OCP
GLOBAL
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October 12-15, 2026
San Jose, California